IT MASTERY
COURSE CATALOGUE

Networking Setup | Cloud Computing
Ethical Hacking | Penetration Testing

DETAILED EDITION
54 Courses - Lesson Notes - Labs - 250+ Q&A

Important Notice: Ethical Use Only

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A. NETWORKING SETUP

Course 1: Networking Fundamentals

Level: Beginner **Duration:** 4 weeks **Modules:** 8

Module 1: What is a network, types (LAN/WAN/MAN/PAN)
A network links devices to share data and resources. LAN is one site, MAN a city, WAN spans regions, PAN is personal.

Module 2: Topologies: bus, star, ring, mesh, hybrid
Topology is the logical or physical layout. Star is most common today; mesh gives redundancy; bus and ring are legacy.

Module 3: OSI model - 7 layers in depth
Seven layers: Physical, Data Link, Network, Transport, Session, Presentation, Application. Each layer serves the one above and uses PDUs: bits, frames, packets, segments.

Module 4: TCP/IP model and encapsulation
TCP/IP has four layers: Link, Internet, Transport, Application. Data is wrapped with headers as it goes down (encapsulation) and unwrapped on arrival.

Module 5: Network devices: hub, switch, router, bridge, modem, AP
Hubs repeat bits; switches forward frames by MAC; routers forward packets by IP between networks; APs bridge Wi-Fi to wired; modems convert signals.

Module 6: Cabling: UTP, fiber, coaxial, connectors
Twisted pair (Cat5e/6/6a) for short runs, fiber for long distance and high speed, coax for cable internet. Know RJ45, LC, SC connectors and 100 m copper limit.

Module 7: Basic troubleshooting: ping, tracert, ipconfig/ifconfig
Use ping for reachability, traceroute for path, ipconfig/ifconfig for addressing, nslookup for DNS. Work bottom-up from cable to application.

Module 8: Lab: build a small office network in Packet Tracer
Build PCs, a switch, a router and a server in Packet Tracer; assign IPs, test ping across subnets and document the topology.

Hands-on Lab: Build a two-subnet office network in Packet Tracer with a router, two switches, 4 PCs and a server; verify with ping and tracert.

Q: Which OSI layer does a switch work at?
A: Layer 2 (Data Link), using MAC addresses; Layer 3 switches also route.

Q: What is encapsulation?
A: Adding headers (and trailers) at each layer as data moves down the stack.

Q: Why is star topology popular?
A: One cable failure affects one device only, and central switches are easy to manage.

Course 2: IP Addressing and Subnetting

Level: Beginner **Duration:** 3 weeks **Modules:** 7

Module 1: Binary and decimal conversion
Binary is base 2; each octet has 8 bits (128,64,32,16,8,4,2,1). Practise converting both directions until instant.

Module 2: IPv4 classes A-E, private and public ranges
Class A 1-126, B 128-191, C 192-223. Private: 10/8, 172.16/12, 192.168/16. 127/8 is loopback, 169.254/16 is APIPA.

Module 3: CIDR notation and subnet masks
CIDR /n means n network bits. Mask 255.255.255.0 is /24. Hosts per subnet = 2^h - 2.

Module 4: Subnetting and VLSM
Split a block into equal subnets by borrowing host bits; VLSM assigns different sizes, largest first, to avoid waste.

Module 5: Supernetting and route summarisation
Combine contiguous networks into one summary route (e.g., four /24s into a /22) to shrink routing tables.

Module 6: IPv6 addressing: types, EUI-64, compression
128-bit addresses in hex. Types: global unicast, link-local fe80::/10, unique local fc00::/7, multicast ff00::/8. Compress zeros with ::.

Module 7: Lab: design an addressing plan for 5 branch offices
Given 10.10.0.0/16, allocate subnets for HQ 500 hosts, branches 100, 50, 25, 10 and point-to-point links.

Hands-on Lab: Design a VLSM plan for 5 branches from 172.16.0.0/16 and verify each network, broadcast and usable range.

Q: How many usable hosts in a /26?
A: 62 (64 addresses minus network and broadcast).

Q: Why VLSM?
A: It sizes each subnet to need and conserves address space.

Q: What is fe80::?
A: IPv6 link-local prefix, auto-configured on every interface.

Course 3: CompTIA Network+ Prep

Level: Beginner **Duration:** 8 weeks **Modules:** 7

Module 1: Networking concepts
Cover ports and protocols (HTTP 80, HTTPS 443, SSH 22, DNS 53, DHCP 67/68), models, cloud and network types.

Module 2: Infrastructure and cabling
Cabling, switching, routing, VLANs, wireless infrastructure, data-centre and virtualisation basics.

Module 3: Network operations and monitoring
Monitoring with SNMP/syslog, backups, change management, disaster recovery, high availability.

Module 4: Network security basics
Threats, attacks, authentication, hardening, firewalls, physical security, remote access.

Module 5: Troubleshooting methodology (7 steps)
CompTIA 7 steps: identify, theorise, test, plan, implement, verify, document.

Module 6: Wireless standards 802.11a/b/g/n/ac/ax
2.4, 5 and 6 GHz bands, channels, MIMO, MU-MIMO, OFDMA, security modes and common interference.

Module 7: Practice exams and exam strategy
Timed practice exams; review wrong answers by objective domain and use performance-based question drills.

Hands-on Lab: Complete two full 90-question mock exams and build a weak-area flashcard deck.

Q: What port does DNS use?
A: UDP/TCP 53.

Q: First step of troubleshooting?
A: Identify the problem by gathering information.

Q: Which Wi-Fi standard is Wi-Fi 6?
A: 802.11ax.

Course 4: Cisco CCNA 200-301

Level: Intermediate **Duration:** 12 weeks **Modules:** 8

Module 1: Network fundamentals
Cover Ethernet, IP, TCP/UDP, cabling, virtualisation and cloud concepts as the exam base.

Module 2: Network access: VLANs, trunking, STP, EtherChannel
VLANs segment broadcast domains; trunks (802.1Q) carry many VLANs; STP prevents loops; EtherChannel bundles links.

Module 3: IP connectivity: static routing, OSPF
Read the routing table, configure static and default routes, and run single-area and multi-area OSPFv2.

Module 4: IP services: NAT, DHCP, DNS, NTP, SNMP, QoS
NAT/PAT, DHCP server and relay, DNS, NTP, SNMP, syslog, QoS markings and per-hop behaviours.

Module 5: Security fundamentals: ACLs, port security, AAA
Standard and extended ACLs, port security, DHCP snooping, DAI, AAA, VPN types, wireless security.

Module 6: Wireless LAN concepts and WLC
Architectures, AP modes, WLC GUI, WLAN creation and WPA2/WPA3 PSK or 802.1X.

Module 7: Automation and programmability: REST, JSON, Ansible
Controller-based vs traditional networking, REST verbs, JSON data, Ansible, Terraform and Cisco DNA Center.

Module 8: Full lab set and mock exams
Run lab sets for every domain and finish with timed mock exams.

Hands-on Lab: Configure a three-router OSPF topology with VLANs, DHCP, NAT and an ACL, and verify with show commands.

Q: What does STP prevent?
A: Layer 2 switching loops and broadcast storms.

Q: Static vs dynamic routing?
A: Static is manual and predictable; dynamic protocols adapt automatically.

Q: What is PAT?
A: NAT overload mapping many private IPs to one public IP using ports.

Course 5: Cisco CCNP Enterprise (ENCOR)

Level: Advanced **Duration:** 12 weeks **Modules:** 8

Module 1: Enterprise architecture and SD-Access
Three-tier and fabric designs, SD-Access, SD-WAN and QoS architecture.

Module 2: Dual-stack routing and virtualisation
Layer 2/3 virtualisation: VRF, GRE, IPsec tunnels, LISP and VXLAN fundamentals.

Module 3: EIGRP and OSPF advanced tuning
Tune EIGRP and OSPF: timers, metrics, summarisation, stub areas and authentication.

Module 4: BGP fundamentals and path selection
BGP neighbours, attributes (weight, local pref, AS path, MED), route maps and filtering.

Module 5: Network assurance, telemetry, NetFlow
Use SNMP, IP SLA, NetFlow, Flexible NetFlow, SPAN and DNA Assurance to prove health.

Module 6: Security: 802.1X, TrustSec, CoPP
802.1X, MAB, TrustSec, CoPP, uRPF and device hardening.

Module 7: Automation: NETCONF, RESTCONF, YANG
Model-driven programmability with YANG, NETCONF, RESTCONF and Python.

Module 8: Capstone enterprise design
Design and implement a full enterprise topology end to end.

Hands-on Lab: Build a dual-stack enterprise with OSPF, BGP to two ISPs, VRF-lite and automated config via Python.

Q: What is the BGP path-selection first tie-breaker (Cisco)?
A: Highest weight (Cisco-local), then local preference.

Q: What is VRF?
A: Multiple isolated routing tables on one router.

Q: YANG purpose?
A: A data-modelling language describing device config and state.

Course 6: Routing Protocols Masterclass

Level: Intermediate **Duration:** 6 weeks **Modules:** 8

Module 1: Static vs dynamic routing
Static routes are manual with low overhead; dynamic routing shares topology automatically; compare administrative distance.

Module 2: RIP v1/v2
RIP uses hop count (max 15); v2 adds VLSM and multicast; suited only for tiny networks.

Module 3: EIGRP metrics and feasibility condition
EIGRP uses bandwidth and delay; the feasibility condition guarantees loop-free backup routes.

Module 4: OSPF areas, LSAs, DR/BDR
OSPF builds a link-state database per area, runs SPF, elects DR/BDR on multi-access links.

Module 5: IS-IS basics
IS-IS runs on Layer 2 using levels 1 and 2; common in service-provider cores.

Module 6: BGP: eBGP, iBGP, attributes, route maps
eBGP between ASes, iBGP within; route maps and prefix lists control policy.

Module 7: Redistribution and route filtering
Redistribute between protocols carefully with metrics and tags; filter to prevent loops.

Module 8: Lab: multi-protocol topology in GNS3
Build a multi-protocol lab in GNS3 and verify with show ip route and debug.

Hands-on Lab: Redistribute EIGRP into OSPF with route tags and prefix-list filtering in GNS3.

Q: Administrative distance of OSPF?
A: 110 (EIGRP internal 90, static 1).

Q: What does the DR do?
A: Reduces adjacency load by acting as hub on multi-access segments.

Q: iBGP split-horizon rule?
A: Routes learned from iBGP peers are not advertised to other iBGP peers.

Course 7: Switching and VLAN Engineering

Level: Intermediate **Duration:** 5 weeks **Modules:** 7

Module 1: Ethernet and MAC learning
Switches learn source MACs, flood unknown unicast and forward known frames from the CAM table.

Module 2: VLANs and 802.1Q trunking
VLANs isolate broadcasts; 802.1Q adds a 4-byte tag; native VLAN is untagged.

Module 3: Inter-VLAN routing (router-on-a-stick, SVI)
Use router-on-a-stick sub-interfaces or SVIs on a Layer 3 switch.

Module 4: Spanning Tree: STP, RSTP, MSTP
STP elects a root bridge; RSTP converges in seconds; MSTP maps VLANs to instances.

Module 5: EtherChannel/LACP
LACP/PAgP bundle ports for bandwidth and redundancy; load-balances by hash.

Module 6: Layer 3 switching
Layer 3 switches route in hardware with ip routing and SVIs.

Module 7: Switch security: DHCP snooping, DAI, port security
Enable port security, DHCP snooping, DAI, BPDU guard and disable unused ports.

Hands-on Lab: Build three VLANs with trunks, RSTP, EtherChannel and port security on three switches.

Q: What is the native VLAN?
A: The VLAN whose frames are sent untagged on a trunk.

Q: What does BPDU guard do?
A: Shuts down an edge port that receives BPDUs.

Q: Why RSTP over STP?
A: Much faster convergence.

Course 8: Wireless Networking and Wi-Fi Design

Level: Intermediate **Duration:** 5 weeks **Modules:** 7

Module 1: RF fundamentals: frequency, channels, interference
Wavelength, frequency, signal strength (dBm), SNR and attenuation determine coverage.

Module 2: 802.11 standards and Wi-Fi 6/6E/7
2.4 GHz has 3 non-overlapping channels (1, 6, 11); 5 and 6 GHz offer more and wider channels; Wi-Fi 6/7 improve density.

Module 3: Site survey and AP placement
Predictive and on-site surveys set AP count, power and placement for target RSSI (about -67 dBm).

Module 4: WLAN controllers, SSIDs, roaming
Controllers centralise management; roaming uses 802.11k/v/r; SSIDs map to VLANs.

Module 5: WPA2/WPA3 and 802.1X enterprise security
WPA3-SAE replaces PSK weaknesses; enterprise uses RADIUS with EAP-TLS or PEAP.

Module 6: Guest portals and captive networks
Captive portals redirect guests; isolate guest VLANs and rate-limit.

Module 7: Troubleshooting Wi-Fi performance
Diagnose co-channel interference, hidden nodes, sticky clients and low data rates.

Hands-on Lab: Perform a mock heat-map survey for a floor plan and justify AP positions and channel plan.

Q: Non-overlapping 2.4 GHz channels?
A: 1, 6 and 11.

Q: WPA3 improvement?
A: SAE handshake and stronger protection against offline guessing.

Q: What is roaming 802.11r?
A: Fast BSS transition for quick handoff.

Course 9: Network Security Fundamentals

Level: Intermediate **Duration:** 6 weeks **Modules:** 8

Module 1: CIA triad and threat landscape
Confidentiality, integrity, availability plus common threats: malware, DoS, MITM, spoofing, insider.

Module 2: Firewalls: packet filter, stateful, NGFW
Packet filters check headers; stateful track sessions; NGFWs add app awareness and IPS.

Module 3: IDS/IPS concepts
IDS detects, IPS blocks; signature and anomaly methods; place sensors at choke points.

Module 4: VPN: IPsec, SSL/TLS, WireGuard
IPsec (IKE, ESP), TLS VPNs and WireGuard secure traffic across untrusted paths.

Module 5: ACLs and segmentation
ACLs filter traffic; VLANs, DMZ and micro-segmentation limit blast radius.

Module 6: NAC and 802.1X
802.1X authenticates devices via RADIUS before granting access.

Module 7: Zero Trust principles
Never trust by location: verify identity, device health and context per request.

Module 8: Security policy and hardening checklists
Write security policies and CIS-based hardening checklists for devices.

Hands-on Lab: Design a segmented network with DMZ, internal and guest zones and write the firewall rule matrix.

Q: Stateful vs stateless firewall?
A: Stateful tracks connection state; stateless inspects each packet alone.

Q: What is a DMZ?
A: A buffer zone hosting public-facing services.

Q: IKE purpose?
A: Negotiates keys and security associations for IPsec.

Course 10: Firewall Administration (Fortinet / Palo Alto / pfSense)

Level: Intermediate **Duration:** 6 weeks **Modules:** 7

Module 1: Firewall architecture and zones
Understand zones, interfaces, trust levels and traffic flow through the device.

Module 2: Policies, NAT and VIPs
Policies are ordered; add source, destination, service, action, logging; NAT/VIP exposes internal servers.

Module 3: Application control and URL filtering
Application ID, URL categories and SSL inspection give granular control.

Module 4: Site-to-site and remote-access VPN
Configure IPsec site-to-site and SSL/remote-access VPN with MFA.

Module 5: High availability
Active-passive HA with state sync gives failover.

Module 6: Logging and SIEM integration
Forward logs to a SIEM; tune alerts and review rule hit counts.

Module 7: Lab: deploy pfSense with VLANs
Deploy pfSense in a VM with WAN, LAN, VLANs, DHCP and rules.

Hands-on Lab: Build pfSense with LAN, DMZ, VLAN and OpenVPN and test rules from each zone.

Q: Why order matters in policies?
A: First match wins.

Q: What is a VIP?
A: Virtual IP mapping a public address to an internal server.

Q: SSL inspection risk?
A: Privacy and certificate-trust issues; exempt sensitive categories.

Course 11: Linux Networking

Level: Intermediate **Duration:** 4 weeks **Modules:** 7

Module 1: ip, ss, nmcli, netplan
ip addr, ip route, ss -tulpn, nmcli and netplan configure and inspect interfaces.

Module 2: Bridges, bonding, VLAN interfaces
Bridges connect segments; bonds aggregate NICs; VLAN sub-interfaces add 802.1Q tags.

Module 3: iptables and nftables
iptables chains (INPUT, FORWARD, OUTPUT) and nftables rulesets provide host firewalls.

Module 4: DNS (BIND/unbound) and DHCP servers
Run unbound/BIND and dnsmasq/isc-dhcp-server for name and address services.

Module 5: Routing and IP forwarding
Enable net.ipv4.ip_forward and add routes to turn Linux into a router.

Module 6: tcpdump and traffic analysis
tcpdump with BPF filters captures traffic for analysis in Wireshark.

Module 7: Network namespaces and containers
Namespaces isolate network stacks; the basis of Docker and Kubernetes networking.

Hands-on Lab: Turn a Linux VM into a router with NAT, DHCP and nftables rules.

Q: Command to list listening ports?
A: ss -tulpn.

Q: What enables routing in Linux?
A: net.ipv4.ip_forward=1.

Q: What is a network namespace?
A: An isolated network stack with its own interfaces and routes.

Course 12: Network Automation with Python and Ansible

Level: Advanced **Duration:** 6 weeks **Modules:** 7

Module 1: Python for network engineers
Variables, loops, functions, files and modules, framed for network tasks.

Module 2: Netmiko, NAPALM, Nornir
Netmiko sends CLI over SSH; NAPALM gives vendor-neutral getters; Nornir runs tasks in parallel.

Module 3: Jinja2 templates
Jinja2 renders configs from variables so one template serves many devices.

Module 4: Ansible network modules and playbooks
Inventories, playbooks, roles and network modules (ios_config, etc.) are idempotent.

Module 5: REST APIs, JSON, YAML
Call REST APIs with requests; parse JSON and YAML.

Module 6: Git and CI/CD for network configs
Store configs in Git; validate and deploy with pipelines.

Module 7: Lab: automate VLAN rollout on 20 switches
Roll out VLANs to 20 switches with templates and a dry-run check.

Hands-on Lab: Write a Python/Netmiko script that backs up configs from 5 devices and diffs them in Git.

Q: What is idempotency?
A: Repeated runs produce the same final state.

Q: Why Jinja2?
A: Separates config logic from data.

Q: NAPALM benefit?
A: Unified API across vendors.

Course 13: SD-WAN and VPN Technologies

Level: Advanced **Duration:** 4 weeks **Modules:** 7

Module 1: WAN technologies: MPLS, leased lines, broadband
Compare MPLS, leased line, broadband and LTE for cost, SLA and reliability.

Module 2: DMVPN and GRE
GRE tunnels any protocol; DMVPN adds dynamic spoke-to-spoke with NHRP and mGRE.

Module 3: IPsec in depth
Phase 1/2 IKE, ESP, transforms, PFS, route-based vs policy-based VPN.

Module 4: SD-WAN architecture (controller, edge, orchestrator)
Separate management, control and data planes with controller, orchestrator and edge devices.

Module 5: Application-aware routing
Steer traffic by app, loss, latency and jitter across multiple links.

Module 6: Cloud on-ramp
Direct cloud access from branches with local breakout.

Module 7: Monitoring and SLAs
Track SLAs, dashboards and alerts.

Hands-on Lab: Build a DMVPN hub-and-two-spoke lab with IPsec and OSPF.

Q: Why SD-WAN?
A: Cheaper, flexible, app-aware multi-link WAN.

Q: What does NHRP do?
A: Resolves tunnel endpoints to NBMA addresses in DMVPN.

Q: GRE vs IPsec?
A: GRE encapsulates; IPsec encrypts.

Course 14: Network Monitoring and Troubleshooting

Level: Intermediate **Duration:** 4 weeks **Modules:** 7

Module 1: Baselines and KPIs
Baselines show normal; track utilisation, latency, errors and availability.

Module 2: SNMP, syslog, NetFlow/sFlow
SNMP polls, traps notify; NetFlow/sFlow show who talks to whom; syslog gives events.

Module 3: Zabbix, Nagios, PRTG, LibreNMS
Compare Zabbix, Nagios, PRTG and LibreNMS for discovery, alerts and graphs.

Module 4: Wireshark analysis
Filter, follow streams, read TCP flags and analyse retransmissions.

Module 5: Latency, jitter, packet loss diagnosis
Measure with ping, mtr, iperf; locate congestion, duplex mismatches and bad cables.

Module 6: Layered troubleshooting approach
Divide and conquer using OSI layers and compare working vs failing paths.

Module 7: Incident documentation
Record cause, fix and prevention in tickets and runbooks.

Hands-on Lab: Deploy Zabbix, monitor 3 devices and diagnose an induced duplex mismatch.

Q: SNMP v3 benefit?
A: Authentication and encryption.

Q: What is jitter?
A: Variation in packet delay.

Q: Tool for throughput testing?
A: iperf3.

Course 15: Network Design and Architecture

Level: Advanced **Duration:** 6 weeks **Modules:** 7

Module 1: Hierarchical design: core, distribution, access
Core aggregates, distribution applies policy, access connects users.

Module 2: Spine-leaf and data-centre fabrics
Leaf-spine gives predictable east-west latency and easy scaling.

Module 3: High availability and redundancy
Remove single points of failure with redundant links, devices, FHRP and power.

Module 4: Capacity planning
Plan from traffic matrices and growth; keep oversubscription ratios reasonable.

Module 5: QoS design for voice and video
Classify and queue voice (EF), video (AF4x) and best effort.

Module 6: IPv6 migration strategy
Dual-stack first, then phase out IPv4 where possible.

Module 7: Documentation and diagram standards
Produce logical and physical diagrams, IP plans and standards.

Hands-on Lab: Produce a full design document and diagram for a 3-floor office with DR site.

Q: Why three tiers?
A: Scalability and clear failure domains.

Q: Which DSCP for voice?
A: EF (46).

Q: What is oversubscription?
A: Ratio of access bandwidth to uplink capacity.

B. CLOUD COMPUTING

Course 16: Cloud Computing Fundamentals

Level: Beginner **Duration:** 3 weeks **Modules:** 7

Module 1: Cloud concepts and history
On-demand, elastic, metered computing delivered over the internet; five NIST traits: self-service, broad access, pooling, elasticity, measured service.

Module 2: IaaS, PaaS, SaaS, FaaS
IaaS gives VMs and networks; PaaS gives a runtime; SaaS gives apps; FaaS runs functions per event.

Module 3: Public, private, hybrid, multi-cloud
Public is shared; private is dedicated; hybrid links both; multi-cloud uses several providers to limit lock-in.

Module 4: Shared responsibility model
Provider secures hardware and facilities; you secure identities, data, configuration and often the OS.

Module 5: Virtualisation and containers overview
Hypervisors split hardware into VMs; containers share a kernel and start in seconds.

Module 6: Cost models and TCO
Pay-as-you-go vs reserved vs spot; compare capex and opex and watch egress fees.

Module 7: Cloud providers comparison
Compare AWS, Azure, GCP and others on services, regions, pricing and compliance.

Hands-on Lab: Create a free-tier account, launch one VM and one storage bucket, then delete everything and note the cost.

Q: What is elasticity?
A: Automatically scaling resources up or down with demand.

Q: IaaS vs PaaS?
A: IaaS manages infrastructure only; PaaS also manages the runtime.

Q: What is vendor lock-in?
A: Dependence on one provider's proprietary services.

Course 17: AWS Cloud Practitioner

Level: Beginner **Duration:** 4 weeks **Modules:** 7

Module 1: AWS global infrastructure
Regions contain Availability Zones (isolated data centres); edge locations serve CloudFront.

Module 2: IAM basics
Users, groups, roles, policies; enable MFA and avoid the root account.

Module 3: EC2, S3, EBS, VPC overview
EC2 compute, S3 object storage, EBS block volumes, VPC private networking.

Module 4: Pricing, billing, support plans
Free tier, on-demand, savings plans, reserved, spot; Budgets and Cost Explorer; support tiers.

Module 5: Well-Architected Framework
Six pillars: operational excellence, security, reliability, performance, cost, sustainability.

Module 6: Security and compliance
Shared responsibility, KMS, Shield, WAF, Artifact for compliance reports.

Module 7: Exam preparation
Practise on official sample questions and timed mocks.

Hands-on Lab: Launch an EC2 instance, attach an S3 bucket with an IAM role, then set a billing alarm.

Q: What is an AZ?
A: One or more isolated data centres inside a region.

Q: Root account best practice?
A: Lock it with MFA and use IAM users or roles daily.

Q: S3 is which storage type?
A: Object storage.

Course 18: AWS Solutions Architect Associate

Level: Intermediate **Duration:** 10 weeks **Modules:** 10

Module 1: IAM, Organizations, SCPs
Organizations group accounts; SCPs set guardrails; IAM policies are evaluated with explicit deny winning.

Module 2: VPC: subnets, route tables, NAT, endpoints, peering, Transit Gateway
Design CIDRs, public/private subnets, IGW, NAT, endpoints, peering and Transit Gateway.

Module 3: EC2, Auto Scaling, ELB
Launch templates, Auto Scaling policies, ALB/NLB, health checks and placement groups.

Module 4: S3, EFS, FSx, storage classes
S3 classes (Standard to Glacier), lifecycle, versioning, replication; EFS shared files; FSx for Windows/Lustre.

Module 5: RDS, Aurora, DynamoDB
Multi-AZ RDS, read replicas, Aurora, DynamoDB capacity and global tables.

Module 6: Route 53, CloudFront
Hosted zones and routing policies; CloudFront caching and OAC.

Module 7: SQS, SNS, EventBridge
Decouple with queues (SQS), pub/sub (SNS) and event buses (EventBridge).

Module 8: Serverless: Lambda, API Gateway
Lambda concurrency, API Gateway stages, Step Functions.

Module 9: Monitoring: CloudWatch, CloudTrail
CloudWatch metrics and alarms; CloudTrail audit logs; Config rules.

Module 10: Mock exams
Take full-length practice exams and review architecture scenarios.

Hands-on Lab: Deploy a 3-tier web app across 2 AZs with ALB, Auto Scaling and Multi-AZ RDS.

Q: Explicit deny vs allow?
A: Explicit deny always overrides allow.

Q: SQS vs SNS?
A: SQS is a queue (pull); SNS is pub/sub (push).

Q: Make RDS highly available?
A: Enable Multi-AZ.

Course 19: AWS SysOps and DevOps Engineer

Level: Advanced **Duration:** 8 weeks **Modules:** 7

Module 1: CloudFormation and CDK
Define stacks as templates; CDK generates them from code; use change sets and drift detection.

Module 2: CodePipeline, CodeBuild, CodeDeploy
Pipelines orchestrate source, build, test and deploy stages with approvals.

Module 3: Systems Manager
Patch, run commands, store parameters and secrets and use Session Manager instead of SSH.

Module 4: Config, Inspector, GuardDuty
Config evaluates rules; Inspector scans workloads; GuardDuty detects threats.

Module 5: Blue/green and canary deployments
Shift traffic gradually using weighted routing, CodeDeploy and rollbacks.

Module 6: Disaster recovery strategies
Define RPO and RTO; backup/restore, pilot light, warm standby, multi-site.

Module 7: Cost optimisation
Rightsize, use Savings Plans, tag resources and alert on anomalies.

Hands-on Lab: Create a CloudFormation stack and a CodePipeline that deploys it with a manual approval.

Q: RPO vs RTO?
A: RPO is acceptable data loss; RTO is acceptable downtime.

Q: Why Session Manager?
A: No open ports or SSH keys needed.

Q: What is drift?
A: Real resources differing from the template.

Course 20: Microsoft Azure Fundamentals (AZ-900)

Level: Beginner **Duration:** 3 weeks **Modules:** 6

Module 1: Azure architecture: regions, subscriptions, resource groups
Regions, availability zones, subscriptions, management groups and resource groups organise resources.

Module 2: Compute, networking, storage services
VMs, App Service, VNets, load balancers, storage accounts, blobs and disks.

Module 3: Identity: Entra ID, RBAC
Entra ID identities, MFA, conditional access and RBAC roles.

Module 4: Governance: Policy, Blueprints, Cost Management
Azure Policy enforces standards; Cost Management tracks spend; tags organise.

Module 5: Security: Defender for Cloud
Defender for Cloud scores posture and recommends fixes.

Module 6: Monitoring: Azure Monitor
Azure Monitor, Log Analytics, Advisor and Service Health.

Hands-on Lab: Create a resource group, a VM and a storage account; assign Reader to a test user.

Q: Resource group purpose?
A: Logical container for related resources.

Q: What is RBAC?
A: Role-based access control assigning permissions at a scope.

Q: Azure Policy does what?
A: Audits and enforces compliance rules.

Course 21: Azure Administrator (AZ-104)

Level: Intermediate **Duration:** 8 weeks **Modules:** 7

Module 1: Entra ID users, groups, RBAC
Manage users, groups, licences, SSPR, and custom roles at various scopes.

Module 2: VMs, scale sets, App Service, containers
Deploy VMs, scale sets, App Service plans, ACI and AKS basics.

Module 3: Virtual networks, NSGs, load balancers, VPN Gateway
VNets, subnets, NSGs, peering, VPN Gateway, Load Balancer and Bastion.

Module 4: Storage accounts, blobs, files
Storage redundancy (LRS/ZRS/GRS), SAS tokens, blob tiers and Azure Files.

Module 5: Backup and Site Recovery
Recovery Services vault, backup policies and Site Recovery replication.

Module 6: Azure Monitor and Log Analytics
Metrics, alerts, action groups and Log Analytics KQL queries.

Module 7: ARM templates and Bicep
Deploy repeatably with ARM templates and Bicep modules.

Hands-on Lab: Deploy a VNet with two subnets, NSGs and a VM scale set using Bicep.

Q: NSG purpose?
A: Filters network traffic by priority-ordered rules.

Q: GRS means?
A: Geo-redundant storage replicated to a paired region.

Q: What is Bicep?
A: A declarative language compiling to ARM templates.

Course 22: Azure Solutions Architect (AZ-305)

Level: Advanced **Duration:** 8 weeks **Modules:** 7

Module 1: Identity and governance design
Design management groups, Policy, PIM, conditional access and hybrid identity.

Module 2: Data storage design
Choose between SQL, Cosmos DB, Blob and Data Lake by access pattern.

Module 3: Business continuity design
Plan backup, replication, failover and availability across zones and regions.

Module 4: Infrastructure and network design
Hub-and-spoke, Front Door, App Gateway, firewall and private endpoints.

Module 5: Hybrid connectivity: ExpressRoute
ExpressRoute and VPN for on-prem links with routing design.

Module 6: Migration strategies
Assess with Azure Migrate; pick rehost, refactor, rearchitect or rebuild.

Module 7: Landing zones
Design Azure landing zones with subscriptions, policy and network baselines.

Hands-on Lab: Draw a hub-and-spoke landing zone with policy, monitoring and DR.

Q: ExpressRoute vs VPN?
A: ExpressRoute is private and predictable; VPN uses the public internet.

Q: What is PIM?
A: Just-in-time privileged access for roles.

Q: Landing zone?
A: Pre-configured environment ready for workloads.

Course 23: Google Cloud Associate Engineer

Level: Intermediate **Duration:** 8 weeks **Modules:** 7

Module 1: Projects, IAM, resource hierarchy
Organisation, folders, projects, IAM roles and service accounts.

Module 2: Compute Engine, GKE, Cloud Run
Compute Engine VMs, GKE clusters and serverless Cloud Run.

Module 3: VPC, firewall, load balancing
Custom VPCs, firewall rules, Cloud NAT, load balancers.

Module 4: Cloud Storage, Cloud SQL, BigQuery
Cloud Storage classes, Cloud SQL, Spanner and BigQuery basics.

Module 5: Cloud Monitoring and Logging
Cloud Logging, Monitoring, Trace and alerting policies.

Module 6: Deployment Manager and Terraform
Infrastructure with Terraform and Deployment Manager.

Module 7: Billing and quotas
Budgets, alerts, quotas and committed use discounts.

Hands-on Lab: Deploy an app on Cloud Run behind a load balancer and monitor it with alerts.

Q: Project in GCP?
A: The base unit for resources, billing and IAM.

Q: Cloud Run is?
A: Serverless container platform.

Q: BigQuery is?
A: Serverless data warehouse.

Course 24: Docker and Containers

Level: Intermediate **Duration:** 4 weeks **Modules:** 7

Module 1: Container concepts vs VMs
Containers share the host kernel and isolate using namespaces and cgroups.

Module 2: Images, Dockerfiles, layers
Dockerfile instructions (FROM, RUN, COPY, CMD); layers cache builds; use multi-stage builds.

Module 3: Volumes and networking
Named volumes persist data; bridge networks connect containers by name.

Module 4: Docker Compose
Describe multi-service apps in compose.yaml and run with docker compose up.

Module 5: Registries and image scanning
Push to registries; scan with Trivy; sign and pin digests.

Module 6: Container security best practices
Run non-root, drop capabilities, read-only filesystems, minimal base images.

Module 7: Lab: containerise a 3-tier app
Containerise a web app, database and cache with Compose.

Hands-on Lab: Containerise a Flask app with PostgreSQL and Redis; scan the image with Trivy.

Q: Image vs container?
A: Image is the template; container is a running instance.

Q: Why multi-stage builds?
A: Smaller, safer final images.

Q: Persist data how?
A: Volumes or bind mounts.

Course 25: Kubernetes Administration (CKA)

Level: Advanced **Duration:** 8 weeks **Modules:** 8

Module 1: Cluster architecture
Control plane: API server, etcd, scheduler, controller manager; nodes run kubelet and runtime.

Module 2: Pods, Deployments, StatefulSets, DaemonSets
Deployments manage replicas and rollouts; StatefulSets keep identity; DaemonSets run per node.

Module 3: Services, Ingress, NetworkPolicy
ClusterIP, NodePort, LoadBalancer, Ingress and default-deny NetworkPolicies.

Module 4: ConfigMaps, Secrets, RBAC
Inject config and secrets; define Roles, RoleBindings and ServiceAccounts.

Module 5: Storage: PV, PVC, StorageClass
PV, PVC, StorageClass and dynamic provisioning.

Module 6: Scheduling, taints, affinity
Use nodeSelector, affinity, taints and tolerations, and resource requests/limits.

Module 7: Cluster install with kubeadm, upgrades, etcd backup
Bootstrap with kubeadm, upgrade versions, back up and restore etcd.

Module 8: Troubleshooting
Debug with kubectl describe, logs, events and exec.

Hands-on Lab: Build a 3-node kubeadm cluster, deploy an app with Ingress and back up etcd.

Q: What stores cluster state?
A: etcd.

Q: Taint vs toleration?
A: Taint repels pods; toleration lets a pod tolerate it.

Q: Rolling update?
A: Gradually replaces pods with zero downtime.

Course 26: Terraform and Infrastructure as Code

Level: Intermediate **Duration:** 5 weeks **Modules:** 7

Module 1: IaC principles
Declarative, version-controlled infrastructure removes manual drift.

Module 2: HCL syntax, providers, resources
Providers expose resources; variables, outputs and locals parameterise code.

Module 3: State management and remote backends
State maps config to reality; remote backends with locking enable teams.

Module 4: Modules and workspaces
Modules package reuse; workspaces separate environments.

Module 5: Multi-cloud deployments
Combine providers to deploy across AWS, Azure and GCP.

Module 6: Policy as code (Sentinel, OPA)
Sentinel or OPA checks plans against guardrails.

Module 7: CI/CD integration
Run fmt, validate, plan in PRs and apply on merge.

Hands-on Lab: Create a reusable VPC module and deploy it to dev and prod workspaces.

Q: terraform plan does what?
A: Previews changes without applying.

Q: Why remote state?
A: Sharing, locking and safety.

Q: What is a module?
A: A reusable group of resources.

Course 27: DevOps and CI/CD Pipelines

Level: Intermediate **Duration:** 6 weeks **Modules:** 7

Module 1: DevOps culture and lifecycle
Culture of collaboration, automation, measurement and fast feedback; DORA metrics.

Module 2: Git and branching strategies
Branching: trunk-based, GitFlow; pull requests and code review.

Module 3: Jenkins, GitHub Actions, GitLab CI
Pipelines as code in Jenkinsfile, GitHub Actions YAML, .gitlab-ci.yml.

Module 4: Build, test, deploy automation
Automate unit, integration and security tests, then deploy via blue/green or canary.

Module 5: Artifact repositories
Nexus, Artifactory and registries version build outputs.

Module 6: Monitoring: Prometheus and Grafana
Prometheus scrapes metrics; Grafana visualises; alert on SLOs.

Module 7: DevSecOps integration
SAST, DAST, SCA and secret scanning inside the pipeline.

Hands-on Lab: Build a GitHub Actions pipeline that tests, scans and deploys a container.

Q: What is CI?
A: Frequent merging with automated build and test.

Q: CD vs CD?
A: Delivery needs approval; deployment is fully automatic.

Q: What is DevSecOps?
A: Security built into every pipeline stage.

Course 28: Cloud Networking and Hybrid Connectivity

Level: Advanced **Duration:** 5 weeks **Modules:** 7

Module 1: Cloud VPC/VNet design
Plan non-overlapping CIDRs, subnet tiers and route tables.

Module 2: Peering, Transit Gateway, hub-and-spoke
Hub-and-spoke centralises shared services; Transit Gateway scales peering.

Module 3: VPN and Direct Connect/ExpressRoute
IPsec VPN for quick links; Direct Connect/ExpressRoute for private bandwidth.

Module 4: DNS in cloud, private endpoints
Private DNS zones and private endpoints keep traffic off the internet.

Module 5: Load balancers and WAF
Layer 7 ALB with WAF; Layer 4 NLB for TCP/UDP.

Module 6: Network security groups and firewalls
SGs are stateful per-instance; NSGs and NACLs filter at subnet level.

Module 7: Multi-region architecture
Replicate across regions with global DNS and traffic managers.

Hands-on Lab: Connect two VPCs and an on-prem simulator with a Transit Gateway and VPN.

Q: Security group vs NACL?
A: SG is stateful at instance; NACL is stateless at subnet.

Q: Why private endpoints?
A: Access PaaS over private IPs.

Q: Overlapping CIDR problem?
A: Prevents peering and routing.

Course 29: Cloud Security (CCSP / Security+ Cloud)

Level: Advanced **Duration:** 8 weeks **Modules:** 7

Module 1: Cloud security architecture
Defence in depth: identity, network, workload, data and monitoring layers.

Module 2: Identity federation, SSO, MFA
Federation with SAML/OIDC, SSO, MFA and short-lived credentials.

Module 3: Data encryption, KMS, HSM
Encrypt at rest and in transit; manage keys with KMS/HSM and rotation.

Module 4: Logging and SIEM in cloud
Collect CloudTrail, Activity Logs and flow logs into a SIEM.

Module 5: CSPM and misconfiguration detection
Detect public buckets, open ports and over-privileged roles continuously.

Module 6: Compliance: ISO 27001, SOC 2, GDPR, HIPAA
Map controls to ISO 27001, SOC 2, GDPR and HIPAA.

Module 7: Incident response in cloud
Prepare, contain, eradicate, recover with cloud-native forensic snapshots.

Hands-on Lab: Run Prowler against a sandbox account and remediate the top 5 findings.

Q: What is CSPM?
A: Continuous detection of cloud misconfigurations.

Q: Customer-managed keys?
A: Keys you control and rotate yourself.

Q: Federation benefit?
A: Central identity with no duplicated passwords.

Course 30: Serverless and Cloud Migration

Level: Intermediate **Duration:** 4 weeks **Modules:** 7

Module 1: Serverless architecture patterns
Event-driven, stateless, pay-per-use compute with auto-scaling.

Module 2: Lambda, Azure Functions, Cloud Functions
Lambda, Azure Functions and Cloud Functions triggers, limits and cold starts.

Module 3: Event-driven design
Events from queues, buckets and streams decouple services.

Module 4: The 6 Rs of migration
Rehost, replatform, repurchase, refactor, retire, retain.

Module 5: Assessment and discovery tools
Discovery tools map dependencies and size workloads.

Module 6: Database migration
DMS and native tools replicate data with minimal downtime.

Module 7: Post-migration optimisation
Rightsize, tag, automate and review costs after cutover.

Hands-on Lab: Build an S3-triggered Lambda that resizes images and logs to CloudWatch.

Q: Cold start?
A: Latency when a new function instance initialises.

Q: Rehost?
A: Lift-and-shift with minimal change.

Q: Why decouple with queues?
A: Resilience and independent scaling.

C. ETHICAL HACKING

Course 31: Introduction to Ethical Hacking

Level: Beginner **Duration:** 4 weeks **Modules:** 6

Module 1: Hacker types: white, grey, black hat
White hats work with permission; black hats act illegally; grey hats sit between. Ethical hackers always have written authorisation.

Module 2: Cyber kill chain and MITRE ATT&CK
Lockheed's kill chain and MITRE ATT&CK describe attacker stages and techniques so defenders can map detections.

Module 3: Legal, ethics and laws (IT Act, CFAA, GDPR)
Know IT Act 2000, CFAA, GDPR; unauthorised access is a crime even with good intent.

Module 4: Setting up a home lab: VirtualBox, Kali, Metasploitable
Install VirtualBox, Kali and intentionally vulnerable VMs on a host-only network isolated from your real LAN.

Module 5: Basic Linux for hackers
Master navigation, grep, permissions, pipes and package management.

Module 6: Reporting and responsible disclosure
Write findings with evidence, impact and fixes; follow coordinated disclosure.

Hands-on Lab: Build an isolated lab with Kali plus Metasploitable and take snapshots.

Q: What authorises a test?
A: Written scope and permission from the asset owner.

Q: Why use host-only networking?
A: Keeps vulnerable VMs off real networks.

Q: What is ATT&CK?
A: A knowledge base of adversary tactics and techniques.

Course 32: CEH v13 Certified Ethical Hacker Prep

Level: Intermediate **Duration:** 12 weeks **Modules:** 10

Module 1: Footprinting and reconnaissance
Passive and active information gathering to map the target footprint.

Module 2: Scanning networks
Identify live hosts, open ports and services using scanning techniques.

Module 3: Enumeration
Pull usernames, shares and service data from exposed services.

Module 4: Vulnerability analysis
Match findings to CVEs, score by CVSS and prioritise.

Module 5: System hacking concepts
Concepts of authentication attacks, privilege escalation and persistence, with matching defences.

Module 6: Malware threats (concepts and defence)
Types of malware, how they spread, and how antivirus, EDR and sandboxing detect them.

Module 7: Sniffing, social engineering, DoS
Sniffing, social engineering, denial of service and their countermeasures.

Module 8: Session hijacking, IDS/firewall evasion concepts
Session hijacking, evasion of IDS/firewalls and how to detect them.

Module 9: Web server and web application hacking
Web server and application attack classes and mitigations.

Module 10: SQL injection, wireless, mobile, IoT, cloud, cryptography
SQLi, wireless, mobile, IoT, cloud and cryptography overviews with defences.

Hands-on Lab: Complete a full simulated engagement against a lab network and write the report.

Q: What does CEH certify?
A: Knowledge of attacker tools and techniques for defensive use.

Q: Why vulnerability analysis?
A: To prioritise real risk.

Q: DoS defence?
A: Rate limiting, CDN/scrubbing, and redundancy.

Course 33: Linux for Hackers

Level: Beginner **Duration:** 3 weeks **Modules:** 7

Module 1: File system and permissions
Everything is a file; know /etc, /var, /home, /tmp, /proc.

Module 2: Bash scripting
Variables, loops, conditionals and functions to automate tasks.

Module 3: Process and service management
ps, top, systemctl, journalctl, cron.

Module 4: Networking commands
ip, ss, curl, wget, nc, dig.

Module 5: Package management
apt, dpkg, pip and building from source safely.

Module 6: Log analysis
Read /var/log/auth.log, syslog and journald for events.

Module 7: Privilege model: sudo, SUID, capabilities
Understand sudoers, SUID bits and capabilities, and how misconfiguration creates risk.

Hands-on Lab: Write a Bash script that summarises failed SSH logins from auth.log.

Q: What does chmod 750 mean?
A: Owner rwx, group r-x, others none.

Q: What is SUID?
A: Run a file with owner privileges.

Q: View service logs?
A: journalctl -u service.

Course 34: Reconnaissance and OSINT

Level: Intermediate **Duration:** 4 weeks **Modules:** 7

Module 1: Passive vs active recon
Passive uses public data without touching targets; active interacts directly.

Module 2: WHOIS, DNS, subdomain discovery
WHOIS ownership, DNS records, certificate transparency and subdomain enumeration.

Module 3: Search-engine dorking
Operators like site:, filetype:, inurl: reveal exposed documents on your own domains.

Module 4: Shodan and Censys
Search internet-connected devices to find exposed services.

Module 5: theHarvester, Maltego, Recon-ng
Aggregate emails, hosts and relationships from public sources.

Module 6: Social media OSINT ethics
Stay inside law and platform terms; avoid personal privacy invasion.

Module 7: Documenting findings
Record sources, timestamps and confidence.

Hands-on Lab: Map the external attack surface of a domain you own and list exposed items.

Q: Passive recon example?
A: Certificate transparency lookup.

Q: What does Shodan index?
A: Banners of internet-facing devices.

Q: Why record sources?
A: Reproducibility and evidence.

Course 35: Network Scanning and Enumeration

Level: Intermediate **Duration:** 4 weeks **Modules:** 7

Module 1: TCP/UDP scanning theory
TCP uses flags SYN/ACK/FIN/RST; UDP has no handshake so scans are slower.

Module 2: Nmap scan types and timing
Choose scan types, timing templates and output formats; -sV and -O add detail.

Module 3: NSE scripts
Scripts check safe discovery items and known vulnerabilities in authorised scope.

Module 4: Service and OS fingerprinting
Banner and TTL clues indicate service and OS.

Module 5: SMB, SNMP, LDAP, DNS enumeration
Enumerate shares, SNMP communities, directory data and zone transfers on lab hosts.

Module 6: Vulnerability scanners: Nessus, OpenVAS
Authenticated scans give accurate patch status.

Module 7: Reading and triaging scan results
Remove false positives, group by root cause and rank by exploitability.

Hands-on Lab: Scan your lab subnet, export XML and produce a ranked findings table.

Q: Why UDP scans are slow?
A: Closed ports return ICMP with rate limits and open ports are silent.

Q: What is a banner?
A: Service self-identification text.

Q: Triaging means?
A: Validate and prioritise findings.

Course 36: Web Application Hacking (OWASP Top 10)

Level: Intermediate **Duration:** 8 weeks **Modules:** 10

Module 1: HTTP, cookies, sessions
HTTP methods, headers, cookies, sessions and statelessness.

Module 2: Broken access control
Missing authorisation checks let users reach others' data; enforce server-side checks and deny by default.

Module 3: Injection: SQL, command, LDAP
Untrusted input interpreted as commands; fix with parameterised queries and validation.

Module 4: XSS: reflected, stored, DOM
Script injection into pages; defend with output encoding and CSP.

Module 5: CSRF and SSRF
CSRF abuses logged-in browsers; SSRF abuses servers; use tokens and allow-lists.

Module 6: Authentication and session flaws
Weak passwords, session fixation, missing MFA; use secure cookies and rotation.

Module 7: Insecure deserialisation, XXE
Unsafe object parsing and XML entity expansion; disable DTDs and avoid untrusted deserialisation.

Module 8: Security misconfiguration
Default credentials, verbose errors, open cloud storage; harden and automate checks.

Module 9: Burp Suite workflow
Proxy, Repeater, Intruder and Decoder for manual testing in lab.

Module 10: Labs: DVWA, Juice Shop, WebGoat
Practise on DVWA, Juice Shop and WebGoat and write fixes.

Hands-on Lab: Find and fix five OWASP Top 10 flaws in Juice Shop and document each.

Q: Best SQLi defence?
A: Parameterised queries.

Q: XSS defence?
A: Output encoding plus CSP.

Q: What is IDOR?
A: Accessing objects by changing identifiers without authorisation checks.

Course 37: Wireless Network Hacking and Defence

Level: Intermediate **Duration:** 4 weeks **Modules:** 6

Module 1: WLAN security protocols WEP/WPA/WPA2/WPA3
Compare protocols; WEP is broken, WPA3 is current best.

Module 2: Monitor mode and capture
Monitor mode lets a card capture all frames for lab analysis.

Module 3: Handshake analysis in lab environments
Study how handshakes work and why long random passphrases resist cracking.

Module 4: Rogue AP and evil twin detection
Detect unauthorised APs by BSSID, location and signal on your own site.

Module 5: Wireless IDS
WIDS and WIPS alert on deauth floods and rogue devices.

Module 6: Hardening enterprise Wi-Fi
Use WPA3/Enterprise, disable WPS, segment guests, monitor.

Hands-on Lab: Audit your own Wi-Fi: capture beacons, review security settings and harden.

Q: Why is WEP unsafe?
A: Weak IV and RC4 flaws allow fast key recovery.

Q: What is evil twin?
A: Rogue AP imitating a legitimate SSID.

Q: Best Wi-Fi hardening?
A: WPA3, strong keys, WPS off.

Course 38: Social Engineering and Phishing Awareness

Level: Beginner **Duration:** 3 weeks **Modules:** 6

Module 1: Psychology of manipulation
Authority, urgency, scarcity, reciprocity and trust drive manipulation.

Module 2: Phishing, vishing, smishing, pretexting
Email, voice, SMS and invented scenarios; learn indicators.

Module 3: Physical security and tailgating
Tailgating, badge cloning and dumpster diving; use access control and shredding.

Module 4: Running authorised phishing simulations (GoPhish)
Run approved simulations with consent from leadership.

Module 5: Awareness training design
Short recurring modules with real examples beat annual slides.

Module 6: Metrics and reporting
Track click, report and repeat-offender rates.

Hands-on Lab: Plan a consented phishing simulation with a report template.

Q: Top indicator of phishing?
A: Urgency plus mismatched sender or link.

Q: Why simulations?
A: Measure and improve user behaviour.

Q: Tailgating defence?
A: Badges, mantraps and guard checks.

Course 39: Cryptography for Security Professionals

Level: Intermediate **Duration:** 4 weeks **Modules:** 7

Module 1: Symmetric vs asymmetric encryption
AES is symmetric and fast; RSA/ECC are asymmetric and solve key exchange.

Module 2: Hashing and digital signatures
SHA-256 gives integrity; digital signatures prove origin.

Module 3: PKI and certificates
Certificates bind identity to public keys via CAs and chains.

Module 4: TLS handshake
Key exchange, cipher suites, certificates and forward secrecy.

Module 5: Common crypto failures
Weak ciphers, reused IVs, hard-coded keys and poor randomness.

Module 6: Password storage and cracking theory
Use Argon2, bcrypt or scrypt with salts; never fast hashes.

Module 7: Tools: OpenSSL, GPG, Hashcat (authorised)
OpenSSL for certs, GPG for signing, Hashcat to audit own hashes.

Hands-on Lab: Create a CA and issue a server certificate with OpenSSL and test TLS.

Q: Why salt passwords?
A: Defeats precomputed tables and equal-hash comparison.

Q: What is forward secrecy?
A: Past sessions stay safe even if the long-term key leaks.

Q: Hash vs encryption?
A: Hash is one-way; encryption is reversible.

Course 40: Mobile and IoT Security

Level: Advanced **Duration:** 5 weeks **Modules:** 7

Module 1: Android and iOS architecture
Sandboxing, permissions, code signing and secure enclaves.

Module 2: OWASP Mobile Top 10
M1-M10 risks like improper credential usage and insecure storage.

Module 3: Static and dynamic app analysis
Inspect manifests, permissions and network behaviour of lab apps.

Module 4: MobSF, Frida basics
MobSF automates static checks; Frida observes runtime in test apps.

Module 5: IoT protocols: MQTT, Zigbee, BLE
MQTT, Zigbee, BLE and CoAP need authentication and encryption.

Module 6: Firmware analysis concepts
Extract and review firmware for hard-coded secrets in devices you own.

Module 7: Securing devices
Signed updates, unique credentials, network segmentation.

Hands-on Lab: Analyse a sample APK with MobSF and report the top 5 weaknesses.

Q: What is certificate pinning?
A: App trusts only a specific cert or key.

Q: MQTT risk?
A: Unauthenticated brokers expose data.

Q: Why signed firmware?
A: Prevents malicious updates.

Course 41: Malware Analysis Fundamentals

Level: Advanced **Duration:** 6 weeks **Modules:** 7

Module 1: Malware families and behaviour
Viruses, worms, trojans, ransomware, RATs and loaders differ by behaviour.

Module 2: Safe analysis lab setup
Use isolated VMs, snapshots, no shared folders and a host-only network.

Module 3: Static analysis: strings, PE headers
Hashes, strings, imports and headers reveal capabilities without running.

Module 4: Dynamic analysis and sandboxes
Observe processes, registry, files and network in a sandbox.

Module 5: Basic reverse engineering with Ghidra
Disassemble to follow logic and decode configs in lab samples.

Module 6: YARA rules
Write signatures that detect families.

Module 7: Writing analysis reports
Summarise IOCs, behaviour and detection advice.

Hands-on Lab: Analyse a benign training sample in a sandbox and write an IOC report.

Q: Static vs dynamic analysis?
A: Static inspects without execution; dynamic runs it.

Q: What is an IOC?
A: Observable evidence of compromise.

Q: Why snapshots?
A: Revert the lab to clean state.

Course 42: Digital Forensics and Incident Response

Level: Advanced **Duration:** 6 weeks **Modules:** 7

Module 1: Forensic principles and chain of custody
Preserve evidence, document handling and use hashes to prove integrity.

Module 2: Disk and memory acquisition
Create forensic images and capture RAM using write blockers.

Module 3: Autopsy, FTK Imager, Volatility
Autopsy, FTK Imager and Volatility parse disks and memory.

Module 4: Windows and Linux artefacts
Registry, event logs, prefetch, browser and shell histories.

Module 5: Log and timeline analysis
Correlate logs to build a timeline of events.

Module 6: IR lifecycle: NIST 800-61
Prepare, detect, contain, eradicate, recover, learn.

Module 7: Reporting for legal use
Clear, factual, reproducible reports suitable for legal review.

Hands-on Lab: Image a USB lab disk and recover artefacts with Autopsy.

Q: Why hash evidence?
A: Prove it hasn't changed.

Q: Volatile first?
A: RAM is lost at shutdown, so capture it early.

Q: IR first step after detection?
A: Contain without destroying evidence.

D. PENETRATION TESTING

Course 43: Penetration Testing Methodology

Level: Intermediate **Duration:** 4 weeks **Modules:** 6

Module 1: PTES, OSSTMM, NIST 800-115
PTES, OSSTMM and NIST SP 800-115 give repeatable structure.

Module 2: Scoping, rules of engagement, contracts
Contracts, scope, IP ranges, timing, safe-handling, emergency contacts.

Module 3: Threat modelling
Identify assets, attackers and attack paths to focus effort.

Module 4: Phases: recon, exploit, post-exploit, report
Recon, vulnerability analysis, exploitation, post-exploitation, reporting.

Module 5: Risk rating with CVSS
CVSS base, temporal and environmental scores plus business context.

Module 6: Client communication
Kick-off, status updates and out-of-band alerts for critical findings.

Hands-on Lab: Write a full scope and rules-of-engagement document for a mock client.

Q: What is scope creep risk?
A: Testing outside the authorised assets.

Q: CVSS purpose?
A: Standard severity scoring.

Q: Why ROE?
A: Defines legal boundaries.

Course 44: eJPT / Junior Penetration Tester

Level: Beginner **Duration:** 6 weeks **Modules:** 7

Module 1: Networking and web basics refresher
Refresh TCP/IP, HTTP and Linux/Windows basics.

Module 2: Host discovery and port scanning
Find hosts and open ports methodically.

Module 3: Vulnerability assessment
Match services to known weaknesses.

Module 4: Metasploit framework basics
Use modules, options and sessions in lab targets.

Module 5: Password attacks in lab
Understand credential weaknesses and policy defences.

Module 6: Pivoting basics
Route traffic through compromised lab hosts to reach internal segments.

Module 7: Exam-style practice labs
Timed practice labs mirroring the exam style.

Hands-on Lab: Complete a multi-host lab and note each step in a journal.

Q: First phase of a test?
A: Information gathering.

Q: Why pivot?
A: Reach networks not directly accessible.

Q: What is a payload?
A: Code delivered by an exploit.

Course 45: CompTIA PenTest+

Level: Intermediate **Duration:** 8 weeks **Modules:** 6

Module 1: Planning and scoping
Governance, compliance and scoping.

Module 2: Information gathering and vulnerability scanning
Combine recon with scanning and analysis.

Module 3: Attacks and exploits
Network, wireless, application and social attacks at concept level.

Module 4: Reporting and communication
Findings, risk ratings, mitigation and debrief.

Module 5: Tools and code analysis
Read scripts, understand tool output and automation.

Module 6: Remediation recommendations
Technical and process fixes for each finding.

Hands-on Lab: Draft a client-ready report from a lab test.

Q: PenTest+ focus?
A: Planning, attacks, reporting.

Q: Good remediation includes?
A: Actionable, prioritised fixes.

Q: Why read code?
A: To spot logic and secrets.

Course 46: OSCP-Style Penetration Testing

Level: Advanced **Duration:** 16 weeks **Modules:** 9

Module 1: Enumeration mastery
Thorough enumeration beats exploitation; never skip a service.

Module 2: Linux privilege escalation
Find misconfigured services, SUID files, cron jobs and weak permissions.

Module 3: Windows privilege escalation
Services, tokens, registry and unquoted paths in lab hosts.

Module 4: Active Directory attacks
Domain enumeration and trust relationships.

Module 5: Buffer overflow concepts
Memory corruption fundamentals in a controlled lab.

Module 6: Web exploitation chains
Chain small flaws into full compromise.

Module 7: Tunnelling and pivoting
SSH, SOCKS and port forwarding for internal reach.

Module 8: Proof of concept and report writing
Screenshots, commands, impact, remediation.

Module 9: Hack The Box and TryHackMe practice
Hack The Box and TryHackMe machines under time limits.

Hands-on Lab: Complete 20 retired HTB boxes with notes and one full practice report.

Q: Why enumerate deeply?
A: Most compromises come from overlooked details.

Q: What is SUID escalation?
A: Abusing misconfigured privileged binaries.

Q: Note-taking tool?
A: Obsidian or CherryTree.

Course 47: Active Directory Penetration Testing

Level: Advanced **Duration:** 6 weeks **Modules:** 7

Module 1: AD architecture: forests, domains, GPO
Forests, domains, OUs, trusts, GPOs, and how authentication flows.

Module 2: Enumeration: BloodHound, LDAP
LDAP queries and BloodHound graphs reveal paths to high privilege.

Module 3: Kerberos attacks (Kerberoasting, AS-REP)
Understand service-ticket abuse and why strong service passwords and AES help.

Module 4: Credential abuse and lateral movement
Reused credentials and weak ACLs enable movement; learn detection.

Module 5: Delegation and ACL abuse
Constrained delegation and ACL rights are common escalation paths.

Module 6: Persistence concepts
Know persistence classes so defenders can hunt them.

Module 7: Detection and hardening
Tiered admin, LAPS, gMSAs, protected users, auditing.

Hands-on Lab: Build a small AD lab and use BloodHound to find and then fix three attack paths.

Q: What is Kerberoasting defence?
A: Long random service passwords and gMSA.

Q: What is LAPS?
A: Unique local admin passwords per machine.

Q: BloodHound shows?
A: Privilege relationships as graphs.

Course 48: Web Application Penetration Testing (BSCP-Style)

Level: Advanced **Duration:** 8 weeks **Modules:** 7

Module 1: Advanced authentication bypass
Test password reset, MFA and session logic flaws.

Module 2: Business logic flaws
Abuse of intended workflows like price or step manipulation.

Module 3: API security testing
Test REST/GraphQL authorisation, rate limits and mass assignment.

Module 4: JWT and OAuth weaknesses
Weak signing, alg confusion and flawed redirect handling.

Module 5: Race conditions and cache poisoning
Concurrency flaws and poisoned caches.

Module 6: Advanced SQLi and XSS
Deep injection testing and context-aware XSS.

Module 7: Burp extensions and automation
Write extensions and macros to speed up repetitive tests.

Hands-on Lab: Complete PortSwigger labs in three categories and write fix advice.

Q: What is mass assignment?
A: Binding client data to protected fields.

Q: JWT risk?
A: Weak secrets or accepting alg none.

Q: Race condition?
A: Outcome depends on request timing.

Course 49: Network Penetration Testing

Level: Intermediate **Duration:** 6 weeks **Modules:** 7

Module 1: External and internal testing
Different views of the same risk: perimeter vs insider.

Module 2: Service exploitation with Metasploit
Verify exploitability on authorised lab hosts.

Module 3: Password attacks and spraying
Test lockout policies and credential reuse.

Module 4: MITM and ARP concepts in lab
Understand Layer 2 attacks to defend with DAI and port security.

Module 5: VLAN hopping and segmentation testing
Verify firewalls and VLAN isolation really block flows.

Module 6: Post-exploitation and data discovery
Locate sensitive data to show impact.

Module 7: Cleanup
Remove tools and accounts and restore state.

Hands-on Lab: Test segmentation between three VLANs and report gaps.

Q: Why internal tests?
A: Assume breach scenarios.

Q: Password spraying defence?
A: Lockout, MFA, monitoring.

Q: Why cleanup?
A: Leave systems as found.

Course 50: Cloud Penetration Testing (AWS/Azure/GCP)

Level: Advanced **Duration:** 6 weeks **Modules:** 7

Module 1: Cloud attack surface
Identity, storage, APIs and metadata are the major surfaces.

Module 2: IAM misconfiguration abuse
Wildcard policies and role chaining enable escalation.

Module 3: S3 and storage exposure
Public buckets and weak ACLs leak data.

Module 4: Metadata service risks
IMDSv2 and SSRF protections reduce credential theft.

Module 5: Container and Kubernetes testing
Pod security, RBAC and exposed dashboards.

Module 6: Tools: ScoutSuite, Pacu, Prowler
Prowler, ScoutSuite and Pacu in your own account.

Module 7: Cloud provider testing policies
Follow each provider's permitted-testing policy.

Hands-on Lab: Audit your own AWS account with Prowler and fix the top findings.

Q: IMDSv2 benefit?
A: Session token requirement blocks simple SSRF theft.

Q: Common cloud flaw?
A: Over-permissive IAM.

Q: Need permission to test?
A: Follow provider policy and customer approval.

Course 51: Red Team Operations

Level: Advanced **Duration:** 8 weeks **Modules:** 7

Module 1: Red vs purple vs blue teaming
Red attacks, blue defends, purple collaborates to improve both.

Module 2: Adversary emulation with ATT&CK
Map real threat-actor behaviour to test controls.

Module 3: C2 framework concepts
Know architecture and operating concepts of C2 for detection.

Module 4: Initial access and OPSEC principles
Phishing, exposed services and credentials; defenders learn OPSEC indicators.

Module 5: Lateral movement and persistence
Study how adversaries move and persist to detect them.

Module 6: Detection engineering feedback
Convert findings into detection rules.

Module 7: Operation reports
Narrative reports with timeline and recommendations.

Hands-on Lab: Write an adversary-emulation plan with detections mapped to ATT&CK.

Q: Purple team goal?
A: Joint improvement of detection and defence.

Q: Why ATT&CK?
A: Common language for TTPs.

Q: Output of red team?
A: Detections and fixes, not just access.

Course 52: Vulnerability Management and Reporting

Level: Intermediate **Duration:** 4 weeks **Modules:** 7

Module 1: Asset inventory
You can't protect unknown assets; use CMDB and discovery.

Module 2: Scanning cadence and tools
Weekly or continuous with credentialed scans.

Module 3: CVSS, EPSS, KEV prioritisation
CVSS plus EPSS and CISA KEV for real-world risk.

Module 4: Patch management
Test, deploy, verify and roll back plans.

Module 5: Writing executive and technical reports
Executive summary, technical detail, evidence, fix steps.

Module 6: Remediation tracking
Tickets with owners and SLAs.

Module 7: Metrics and dashboards
MTTR, coverage and aging dashboards.

Hands-on Lab: Write a two-page executive and a technical report for a sample scan.

Q: What is KEV?
A: CISA catalogue of exploited vulnerabilities.

Q: EPSS?
A: Probability of exploitation.

Q: Key metric?
A: Mean time to remediate.

Course 53: Bug Bounty Hunting

Level: Intermediate **Duration:** 6 weeks **Modules:** 7

Module 1: Platforms: HackerOne, Bugcrowd
HackerOne and Bugcrowd list programs, rules and rewards.

Module 2: Scope reading and legal safe harbour
Only test in-scope assets and follow safe-harbour language.

Module 3: Recon automation
Automate asset discovery within scope.

Module 4: Common high-impact bug classes
Access control, injection, logic and exposure issues.

Module 5: Writing quality reports
Clear title, steps, impact, proof, fix.

Module 6: Triage and communication
Be polite, patient and provide extra info.

Module 7: Ethics and responsible disclosure
Never access others' data beyond proof.

Hands-on Lab: Write a sample high-quality report for a Juice Shop flaw.

Q: Why read scope?
A: Out-of-scope testing may be illegal.

Q: Good report has?
A: Reproducible steps and impact.

Q: Duplicate means?
A: Already reported by someone else.

Course 54: Blue Team and SOC Analyst

Level: Intermediate **Duration:** 8 weeks **Modules:** 7

Module 1: SOC structure and tiers
Tier 1 triage, Tier 2 investigation, Tier 3 hunting and engineering.

Module 2: SIEM: Splunk, Elastic, Wazuh
Splunk, Elastic and Wazuh ingest, correlate and alert.

Module 3: Log sources and correlation
Windows events, Sysmon, firewall, proxy, DNS and cloud logs.

Module 4: Threat hunting
Hypothesis-driven search for hidden activity.

Module 5: Sigma and detection rules
Vendor-neutral detection rules shared across SIEMs.

Module 6: Incident handling playbooks
Step-by-step handling of phishing, malware and account compromise.

Module 7: Threat intelligence
Feeds and enrichment give context to alerts.

Hands-on Lab: Deploy Wazuh, ingest Windows logs and write 3 detections.

Q: What is Sysmon?
A: Windows service that logs detailed system activity.

Q: Tier 1 role?
A: Alert triage.

Q: Why Sigma?
A: Portable detection rules.

E. TOOLS - QUESTION AND ANSWER (FULL DETAILS)

Tool 1: Wireshark

Category: Network Analysis

Overview: Open-source packet analyser that captures and dissects traffic on a network interface in real time.

Usage / Examples:

```
Capture filter: port 80 | Display filters: ip.addr==192.168.1.10, tcp.flags.syn==1,
http.request, dns
```

Q: What is Wireshark used for?

A: Troubleshooting network issues, analysing protocols, finding latency or retransmissions, and investigating suspicious traffic.

Q: Capture filter vs display filter?

A: Capture filters (BPF syntax) limit what is recorded; display filters hide packets already captured and are more flexible.

Q: Can it decrypt HTTPS?

A: Only if you supply the TLS session keys (SSLKEYLOGFILE) or the server private key for non-forward-secret suites.

Tool 2: Nmap

Category: Scanning

Overview: Network mapper for host discovery, port scanning, service/version detection and OS fingerprinting.

Usage / Examples:

```
nmap -sn 192.168.1.0/24 | nmap -sV -p 1-1000 target | nmap -A target | nmap --script vuln
target
```

Q: What does -sV do?

A: Probes open ports to determine service name and version.

Q: Difference between -sS and -sT?

A: -sS is a half-open SYN scan needing raw-socket privileges; -sT completes the full TCP handshake using the OS connect() call.

Q: What is NSE?

A: The Nmap Scripting Engine runs Lua scripts for discovery, vulnerability checks and brute-force audits, only on systems you are authorised to test.

Tool 3: Kali Linux

Category: Platform

Overview: Debian-based distribution maintained by Offensive Security that bundles hundreds of security tools.

Usage / Examples:

```
Install as VM or WSL; update with: sudo apt update && sudo apt full-upgrade
```

Q: Is Kali good for beginners?

A: It is for security learning, but not a general desktop OS; learn Linux basics first.

Q: Why run it in a VM?

A: Isolation, snapshots, and safety for lab experiments.

Q: Alternatives?

A: Parrot OS, BlackArch, or a custom toolset on Ubuntu.

Tool 4: Metasploit Framework

Category: Exploitation Framework

Overview: Modular framework for developing, testing and executing exploit modules against authorised targets.

Usage / Examples:

```
msfconsole | search type:exploit name:smb | use <module> | show options | set RHOSTS <ip> | run
```

Q: What are modules?

A: Exploits, auxiliary scanners, payloads, encoders, post-exploitation and evasion modules.

Q: What is Meterpreter?

A: An advanced in-memory payload giving a flexible post-exploitation session.

Q: Is it legal?

A: Yes when used with written permission on systems you own or are contracted to test.

Tool 5: Burp Suite

Category: Web Testing

Overview: Intercepting proxy and toolkit for testing web application security.

Usage / Examples:

```
Proxy, Repeater, Intruder, Scanner (Pro), Decoder, Comparer, Collaborator
```

Q: How do I start?

A: Configure the browser proxy to 127.0.0.1:8080 and install Burp's CA certificate.

Q: Repeater vs Intruder?

A: Repeater manually resends one modified request; Intruder automates many payload variations.

Q: Community vs Professional?

A: Pro adds the automated scanner, saved projects and no throttling of Intruder.

Tool 6: OWASP ZAP

Category: Web Testing

Overview: Free open-source web app scanner and intercepting proxy from OWASP.

Usage / Examples:

```
zap-baseline.py -t https://test.example | Active scan, spider, AJAX spider, API scan
```

Q: ZAP vs Burp?

A: ZAP is free and automation-friendly; Burp has a more polished manual workflow.

Q: Passive vs active scan?

A: Passive only observes traffic; active sends attack payloads and must be authorised.

Q: Use in CI/CD?

A: Yes, via Docker images and baseline scan scripts.

Tool 7: Nessus / OpenVAS

Category: Vulnerability Scanning

Overview: Scanners that check hosts for known vulnerabilities, misconfigurations and missing patches.

Usage / Examples:

```
Create scan template, set targets and credentials, schedule, export report
```

Q: Credentialed vs non-credentialed?

A: Credentialed scans log in and give deeper, more accurate patch data.

Q: False positives?

A: Verify manually and tune plugins; never report unverified findings as fact.

Q: OpenVAS?

A: The open-source engine within Greenbone Vulnerability Management.

Tool 8: John the Ripper and Hashcat

Category: Password Auditing

Overview: Offline password-hash auditing tools used to test password strength in authorised assessments.

Usage / Examples:

```
john --wordlist=list.txt hashes.txt | hashcat -m 0 -a 0 hashes.txt list.txt
```

Q: Why audit password hashes?
A: To prove weak passwords exist and drive policy such as length, MFA and breached-password checks.

Q: CPU vs GPU?
A: Hashcat uses GPUs for far higher speed on fast hashes.

Q: How to defend?
A: Use slow salted hashes (bcrypt, scrypt, Argon2), long passphrases and MFA.

Tool 9: Hydra

Category: Credential Testing

Overview: Online login tester supporting many protocols, used to verify account-lockout and weak-credential exposure.

Usage / Examples:

```
hydra -L users.txt -P pass.txt ssh://target (authorised lab only)
```

Q: Risk of use?
A: Can lock accounts and trigger alerts; always coordinate with the system owner.

Q: Defences?
A: Lockout thresholds, rate limiting, MFA, fail2ban.

Q: Alternatives?
A: Medusa, Ncrack, CrackMapExec for AD contexts.

Tool 10: Aircrack-ng Suite

Category: Wireless

Overview: Set of tools for auditing Wi-Fi security on networks you own.

Usage / Examples:

```
airmon-ng, airodump-ng, aireplay-ng, aircrack-ng
```

Q: What does it assess?
A: Strength of WEP/WPA/WPA2 pre-shared keys and wireless configuration.

Q: Why monitor mode?
A: It lets the adapter capture frames not addressed to it.

Q: Defence?
A: Use WPA3 or WPA2-Enterprise with long, random keys.

Tool 11: Netcat

Category: Networking Utility

Overview: The 'Swiss-army knife' for reading and writing data over TCP/UDP.

Usage / Examples:

```
nc -lvnp 4444 | nc host 80 | nc -zv host 20-100
```

Q: Common legit uses?
A: Port checks, file transfers, banner grabbing, debugging services.

Q: Ncat?
A: Nmap's modern rewrite with SSL and access-control options.

Q: Is it dangerous?
A: It is neutral, but its misuse is flagged by security tools.

Tool 12: tcpdump

Category: Network Analysis

Overview: Command-line packet capture tool.

Usage / Examples:

```
tcpdump -i eth0 port 53 | tcpdump -nn -w out.pcap host 10.0.0.5
```

Q: -n flag?
A: Disables name resolution for faster, cleaner output.

Q: Read a file?
A: tcpdump -r out.pcap.

Q: Why use it over Wireshark?
A: Works on headless servers over SSH.

Tool 13: Cisco Packet Tracer

Category: Simulation

Overview: Free network simulator for building and testing Cisco topologies.

Usage / Examples:

```
Drag devices, connect cables, configure IOS CLI, simulation mode
```

Q: Who is it for?
A: CCNA students and beginners.

Q: Limits?
A: Not all IOS features are supported; use GNS3/EVE-NG for real images.

Q: Where to get it?
A: Free via Cisco Networking Academy.

Tool 14: GNS3 / EVE-NG

Category: Simulation

Overview: Emulators that run real router, switch and firewall images in virtual labs.

Usage / Examples:

```
Import images, build topology, connect to VMs
```

Q: Difference from Packet Tracer?
A: They emulate actual OS images, giving realistic behaviour.

Q: Hardware needs?
A: 16 GB+ RAM recommended for larger labs.

Q: Licensing?
A: You must supply legitimately licensed vendor images.

Tool 15: SolarWinds / Zabbix / PRTG

Category: Monitoring

Overview: Network monitoring platforms for availability, performance and alerting.

Usage / Examples:

```
SNMP polling, traps, dashboards, thresholds
```

Q: What is SNMP?
A: A protocol to query and manage devices; v3 adds authentication and encryption.

Q: Agent vs agentless?
A: Agents give deeper host metrics; agentless uses SNMP, WMI or ICMP.

Q: Free option?
A: Zabbix and LibreNMS are open source.

Tool 16: pfSense / OPNsense

Category: Firewall

Overview: Open-source firewall/router distributions.

Usage / Examples:

```
Interfaces, rules, NAT, VPN (OpenVPN, WireGuard, IPsec), Suricata/Snort packages
```

Q: Why use in labs?
A: Realistic segmentation and VPN practice at no cost.

Q: Rule order?
A: Rules are evaluated top-down; first match wins.

Q: HA?
A: CARP and pfsync provide failover.

Tool 17: Snort / Suricata

Category: IDS/IPS

Overview: Network intrusion detection and prevention engines using signature rules.

Usage / Examples:

```
alert tcp any any -> any 80 (msg:'test'; sid:1000001;)
```

Q: IDS vs IPS?
A: IDS alerts; IPS can block inline.

Q: Suricata advantage?
A: Multi-threading, TLS logging, EVE JSON output.

Q: Rules source?
A: Emerging Threats, Snort community rules.

Tool 18: Splunk / ELK / Wazuh

Category: SIEM

Overview: Log aggregation and correlation platforms for security monitoring.

Usage / Examples:

```
Ingest logs, build searches, dashboards, alerts
```

Q: What is a SIEM?
A: A system that centralises logs, correlates events and raises alerts.

Q: Open-source choice?
A: Wazuh with Elastic or OpenSearch.

Q: Key skill?
A: Writing effective queries and detection rules.

Tool 19: Terraform

Category: Cloud IaC

Overview: Declarative infrastructure-as-code tool for multi-cloud provisioning.

Usage / Examples:

```
terraform init | plan | apply | destroy
```

Q: What is state?
A: A file mapping config to real resources; store remotely with locking.

Q: plan vs apply?
A: plan previews changes; apply executes them.

Q: Modules?
A: Reusable packages of resources.

Tool 20: Ansible

Category: Automation

Overview: Agentless configuration management using YAML playbooks over SSH.

Usage / Examples:

```
ansible all -m ping | ansible-playbook site.yml
```

Q: Idempotent?
A: Running twice leaves the same end state.

Q: Inventory?
A: A list of hosts and groups.

Q: Network use?
A: Modules for Cisco, Juniper, Arista, Fortinet.

Tool 21: Docker

Category: Containers

Overview: Platform for building and running containers.

Usage / Examples:

```
docker build -t app . | docker run -d -p 80:80 app | docker ps
```

Q: Image vs container?
A: An image is a template; a container is a running instance.

Q: Security tip?
A: Run as non-root, scan images, pin versions.

Q: Compose?
A: Defines multi-container apps in YAML.

Tool 22: Kubernetes (kubectl)

Category: Containers

Overview: Container orchestration platform.

Usage / Examples:

```
kubectl get pods | kubectl apply -f app.yaml | kubectl logs pod
```

Q: Pod?
A: Smallest deployable unit with one or more containers.

Q: Service types?
A: ClusterIP, NodePort, LoadBalancer.

Q: RBAC?
A: Controls who can do what in the cluster.

Tool 23: AWS CLI / Azure CLI / gcloud

Category: Cloud CLI

Overview: Command-line interfaces to manage cloud resources.

Usage / Examples:

```
aws s3 ls | az vm list | gcloud compute instances list
```

Q: Authentication?
A: Use SSO or short-lived credentials rather than long-lived keys.

Q: Profiles?
A: Named credential sets, e.g., aws --profile dev.

Q: Scripting?
A: Combine with jq and shell for automation.

Tool 24: Prowler / ScoutSuite

Category: Cloud Security

Overview: Open-source cloud security posture auditing tools.

Usage / Examples:

```
prowler aws | scout aws
```

Q: What do they check?
A: Misconfigurations against CIS benchmarks and best practice.

Q: Permissions needed?
A: Read-only audit roles.

Q: Output?
A: HTML, JSON, CSV reports.

Tool 25: BloodHound

Category: Active Directory

Overview: Graph-based tool that maps AD privilege relationships; used by both testers and defenders.

Usage / Examples:

```
SharpHound collector -> import into BloodHound -> run path queries
```

Q: Why valuable to defenders?
A: Reveals risky attack paths so you can remove them.

Q: Data source?
A: LDAP and session information from AD.

Q: Best practice?
A: Run only with authorisation.

Tool 26: Maltego / theHarvester / Shodan

Category: OSINT

Overview: Tools for gathering public information about organisations and exposed assets.

Usage / Examples:

```
theHarvester -d example.com -b all | Shodan search: org:"Example"
```

Q: Passive?
A: Mostly yes; they use public data sources.

Q: Privacy?
A: Respect laws and scope.

Q: Use?
A: Map your own attack surface.

Tool 27: Ghidra / IDA / Volatility

Category: Reverse Engineering and Forensics

Overview: Tools for disassembling binaries and analysing memory dumps.

Usage / Examples:

```
Ghidra: import binary, auto-analyse | volatility -f mem.raw windows.pslist
```

Q: Use?
A: Malware analysis and incident response.

Q: Ghidra?
A: NSA's free reverse-engineering suite.

Q: Volatility?
A: Extracts processes, connections, and artefacts from RAM.

Tool 28: Git and GitHub Actions

Category: DevOps

Overview: Version control and CI automation.

Usage / Examples:

```
git commit | git push | workflow YAML in .github/workflows
```

Q: Why for network engineers?
A: Track config changes and enable review.

Q: Secrets?
A: Store in encrypted secrets, never in the repo.

Q: Branching?
A: Feature branches and pull requests.

Tool 29: Nikto

Category: Web Scanning

Overview: Open-source web-server scanner that checks for outdated software, dangerous files and misconfigurations.

Usage / Examples:

```
nikto -h https://test.example (only on authorised targets)
```

Q: What does it find?
A: Default files, outdated server versions and risky HTTP settings.

Q: Is it stealthy?
A: No; it is noisy and easily logged.

Q: Next step after a finding?
A: Verify manually and fix configuration.

Tool 30: Gobuster / ffuf

Category: Content Discovery

Overview: Tools that discover hidden directories and files on web servers using wordlists.

Usage / Examples:

```
gobuster dir -u https://test.example -w wordlist.txt
```

Q: Purpose?
A: Find unlinked content such as admin panels or backups on your own sites.

Q: Defence?
A: Remove unused files, restrict admin paths, monitor 404 spikes.

Q: Wordlists?
A: SecLists is the common collection.

Tool 31: sqlmap

Category: Web Testing

Overview: Automated SQL injection detection and testing tool for authorised targets.

Usage / Examples:

```
sqlmap -u 'http://lab/page?id=1' --batch (lab only)
```

Q: Why use it?
A: Confirms whether an injection point exists and shows impact.

Q: Safer settings?
A: Use low risk/level and a test database.

Q: Fix?
A: Parameterised queries and least-privilege DB accounts.

Tool 32: Trivy

Category: Container Security

Overview: Scanner for container images, file systems, IaC and dependencies.

Usage / Examples:

```
trivy image nginx:latest | trivy fs . | trivy config .
```

Q: What does it detect?
A: Known CVEs, secrets and misconfigurations.

Q: CI use?
A: Fail the build on high severity findings.

Q: Output formats?
A: Table, JSON, SARIF.

Tool 33: CyberChef

Category: Utility

Overview: Browser-based 'Swiss-army knife' for encoding, decoding and data transformation.

Usage / Examples:

```
Drag operations: From Base64, URL Decode, XOR, Hash
```

Q: Use?
A: Quickly decode data during analysis.

Q: Runs offline?
A: Yes, as a local HTML app.

Q: Good for?
A: Forensics, CTFs and log cleanup.

Tool 34: Zeek

Category: Network Monitoring

Overview: Network analysis framework producing rich logs of connections, DNS, HTTP and TLS.

Usage / Examples:

```
zeek -r capture.pcap ; review conn.log, dns.log
```

Q: Zeek vs Snort?
A: Zeek creates behaviour logs; Snort matches signatures.

Q: Best use?
A: Threat hunting and forensics.

Q: Log format?
A: TSV or JSON.

Tool 35: Autopsy / FTK Imager

Category: Forensics

Overview: Disk imaging and analysis suites for evidence collection.

Usage / Examples:

```
Create case, add disk image, run ingest modules, review timeline
```

Q: Write blocker?
A: Prevents changes to source media.

Q: Image format?
A: E01 or raw dd.

Q: Hash check?
A: MD5/SHA-256 before and after.

F. GENERAL INTERVIEW AND EXAM Q&A

Q1: What is the difference between a vulnerability, a threat and a risk?
A: A vulnerability is a weakness, a threat is something that can exploit it, and risk is the likelihood and impact of that exploitation.

Q2: What is penetration testing vs vulnerability assessment?
A: An assessment identifies and ranks weaknesses; a penetration test attempts controlled exploitation to demonstrate real impact.

Q3: What are black, grey and white box tests?
A: Black box: no prior knowledge. Grey: partial knowledge. White: full knowledge including source and architecture.

Q4: What is the OSI model?
A: A 7-layer reference: Physical, Data Link, Network, Transport, Session, Presentation, Application.

Q5: TCP vs UDP?
A: TCP is connection-oriented and reliable; UDP is connectionless and faster with no delivery guarantee.

Q6: What is a three-way handshake?
A: SYN, SYN-ACK, ACK to establish a TCP connection.

Q7: What is NAT?
A: Translating private addresses to public ones so multiple hosts share an IP.

Q8: What is a VPN?
A: An encrypted tunnel across an untrusted network.

Q9: What is zero trust?
A: Never trust by default; verify every user, device and request.

Q10: What is the shared responsibility model?
A: The provider secures the cloud; the customer secures what they put in the cloud.

Q11: What is IAM?
A: Identity and Access Management: who can access which resource and what they can do.

Q12: What is least privilege?
A: Granting only the access needed to perform a job.

Q13: What is SQL injection?
A: Untrusted input altering database queries; prevented with parameterised queries.

Q14: What is XSS?
A: Injecting scripts into pages viewed by others; prevented with output encoding and CSP.

Q15: What is MFA?
A: Requiring two or more factors: know, have, are.

Q16: What is a rules-of-engagement document?
A: It defines scope, timing, permitted techniques and contacts for a test.

Q17: Which certifications should I plan?
A: Network+, CCNA, Security+, AWS/Azure associate, CEH, PenTest+, then OSCP/CCNP/CKA.

Q18: How do I practise legally?
A: Use your own lab, TryHackMe, Hack The Box, PortSwigger Academy, OWASP Juice Shop, and bug bounty programs within scope.

G. SUGGESTED LEARNING ROADMAP

Months 1-3	Networking fundamentals, IP and subnetting, Linux basics, Network+
Months 4-6	CCNA, switching and routing labs, firewalls, cloud fundamentals
Months 7-9	Security+, AWS/Azure associate, Docker, scripting with Python
Months 10-12	CEH or PenTest+, web app hacking labs, Kubernetes, Terraform
Year 2	OSCP, AD pentesting, cloud security, red team, specialise